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Soil Hydrology Analyses Report

Report# 052525HG rev A

Summary

This report is for a single soil depth with a total of six soil samples, two larger 401 cc and four 21 cc. The samples were tested at pressures ranging from 0 to 15 bar ASTM D6836-16. The water retention parameters are provided using RETC (USDA's soil water retention curve software). An assessment of the soil moisture calibration of the HydraProbe soil sensor is included. The raw data is provided.

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Section 1 General Information

Customer/Stakeholder	GEI Consultants 11010 White Rock Road, Suite 200 Rancho Cordova, CA 95670 California Department of Water Resources California Cooperative Snow Surveys (CCSS) program
Site Name	Big Meadows (BIM) Snow Survey Site California,
Sample Name	BIM_50_Horz
Sample Info	Horizontal Boring Sample
Depth	50 cm
Date Submitted	May 20, 2025
Laboratory	Stevens Water pilot soil lab in Portland Oregon
Method	ASTM D6836-16
Other Notes	Note that the factory calibration was accurate in the testing at 0% and at saturation for both 401 cc samples. It deviates for mid-range soil moistures in both 401 cc samples. Sample texture was gravel.

Section 2 Results Summary

Results recommended for depth at site.

Sample	Saturation from Tempe Cell (wfv)	Dry Bulk Density (g/cc)
Average	0.324* Best	1.515

See Section 4, Source, Average Saturation is 0.29 of two Samples in Tempe Cell. The H Sample was unusually low because of rocks. Average dry bulk density

Sample	Field Capacity, Soil Moisture (wfv) at 0.33 Bar	Permanent Wilting Point, Soil Moisture (wfv) at 15 Bar
Average	0.0165	0.01145

See Section 4, Average of two samples from modeled Retention Curve. This sample was gravel with almost no water holding capacity.

Van Genuchten Parameters best for depth

Sample	Alpha (cm)	θ_r	θ_s	n	R^2
Average	0.0753	0.0113	0.324	2.403	0.925 to 0.988

See Section 4, Source, Average out of 4 trials.

Depth Specific Calibration for Site

Algorithm	Polynomial	N	R Squared	RMSE WFV
Average Coefficients	$\theta = -0.2143 + 0.09196\varepsilon_r$ $-0.006108\varepsilon_r^2 + 0.000135\varepsilon_r^3$	53	0.917	0.019
Composite Regression	$\theta = -0.1925 + 0.08362\varepsilon_r$ $-0.00531\varepsilon_r^2 + 0.00113\varepsilon_r^3$	53	0.9199	0.019

See Section 5, 3rd order regression. Sources: Average Coefficients from separate regressions of samples and composite regression two combined sample data sets. Where ε_r is the real dielectric permittivity (parameter 6 on HydraProbe Soil Sensor)

Figure 2.1, Averaged Coefficients from separate Regressions

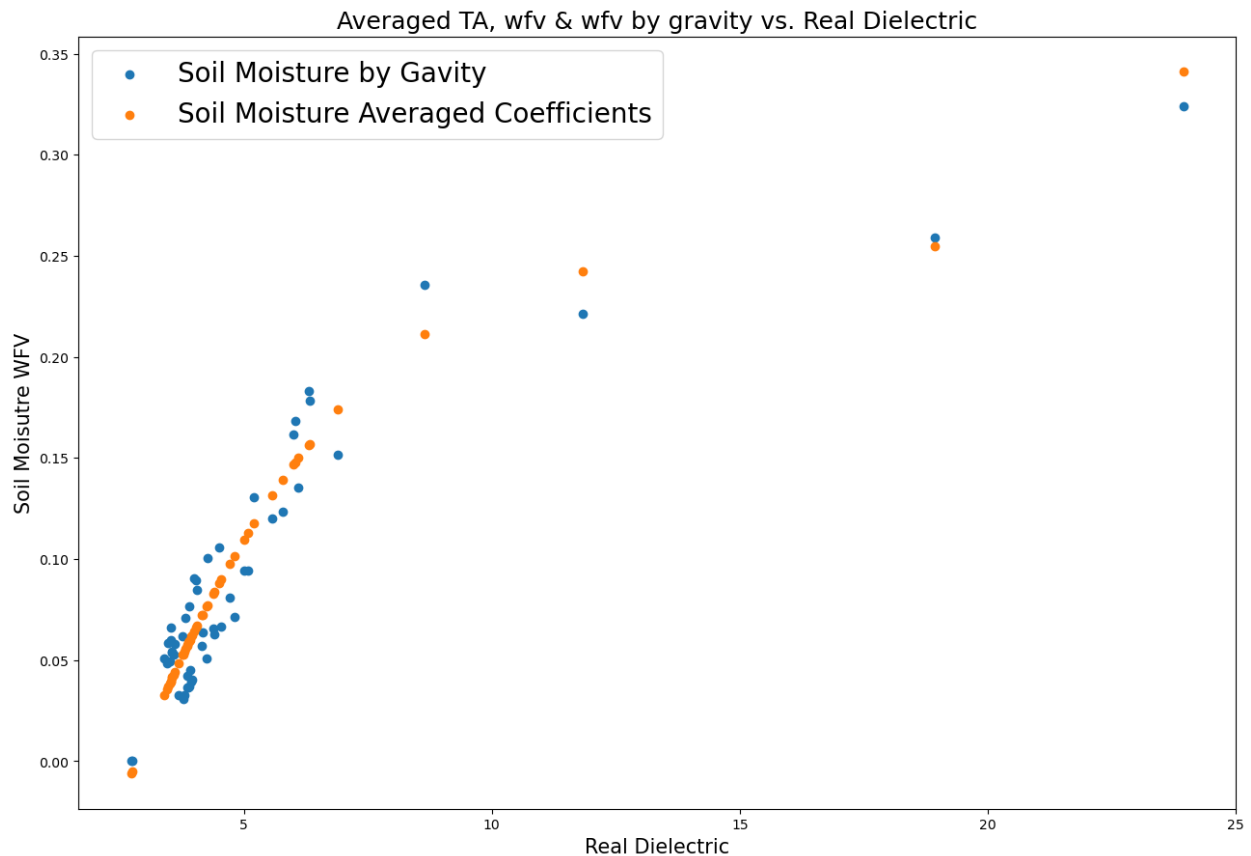
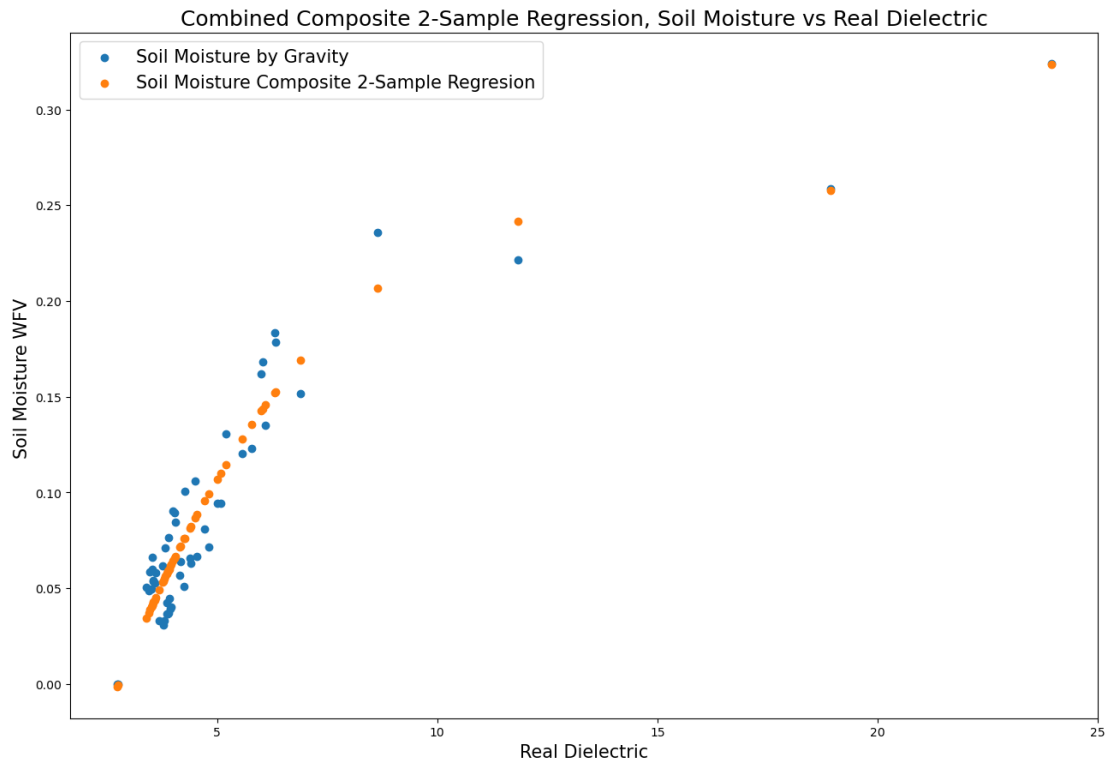


Figure 2.2, Composite Regression



Section 3 Sample Description

Table 3.0, Sample Description for Depth

Sample ID	Remarks	Soil Volume in cc	Sample Type	Date Received
G297541	Sample in a Travel Assy	401	Undisturbed/ partially constructed	5/25/25
H297546	Sample in a Travel Assy	401	Undisturbed/ Partially constructed	5/25/25
G_control1 (G1)	Sample Retaining Ring (1cm/2.25 " OD	20.83	Constructed	5/25/25
G_control2 (G2)	Sample Retaining Ring (1cm/2.25 " OD	20.83	Constructed	5/25/25
H_control1 (H1)	Sample Retaining Ring (1cm/2.25 " OD	20.83	Constructed	5/25/25
H_control2 (H2)	Sample Retaining Ring (1cm/2.25 " OD	20.83	Constructed	5/25/25

Section 4 Results Saturation RETC Model

The measured pressure and moisture data were inputs into the RETC Software to determine the water retention curve parameters using the van Genuchten/Mualem Model ($m = 1-1/n$) with units of hours and cm. More information about RETC software can be available from the US Department of Agriculture ARS,

<https://www.ars.usda.gov/pacific-west-area/riverside-ca/agricultural-water-efficiency-and-salinity-research-unit/docs/model/retc-model/>

$$\theta = \theta_r + \frac{(\theta_s - \theta_r)}{[1 + (\alpha h)^n]^m}$$

The transformation from unsaturated hydraulic head to volumetric water content is described by the van Genuchten Equation where θ_s is the soil's saturation, θ_r is the soil water residual, α is the air entry coefficient and h is unsaturated hydraulic head.

Table 4.1, General Soil Results for bulk density and saturation point.

Sample	Saturation from Tempe Cell (wfv)	Date	Dry Bulk Density (g/cc)	Remarks
G297541	0.324	4/29/26	1.4189	Gravel
H297546	0.259	4/29/26	1.611	Gravel
Average	0.29	-	1.515	-

Table 4.2, Van Genuchten Parameters RETC output.

Sample	Alpha (cm ⁻¹)	θ_r (wfv)	θ_s (wfv)	n	R ²	RMSE (wfv)	N, Number of Samples
G_control1	0.06743	0.00797	0.324	2.41	0.975	0.01	16
G_control2	0.06543	0.00786	0.324	3.044	0.989	0.01	16
H_control1	0.08028	0.0145	0.324*	2.093	0.926	0.02	16
H_control2	0.08805	0.015	0.324*	2.0677	0.950	0.015	18

*A saturation of 0.324 was used because there was a rock in sample H297546

Table 4.3, Field Capacity and Permanent Wilting Points

Sample	Field Capacity, Soil Moisture (wfv) at 0.33 Bar	Permanent Wilting Point, Soil Moisture (wfv) at 15 Bar	Data Source
G_control1	0.01	0.008	Regression from Van Genuchten Parameters
G_control2	0.01	0.008	Regression from Van Genuchten Parameters
H_control1	0.023	0.0146	Regression from Van Genuchten Parameters
H_control2	0.023	0.0152	Regression from Van Genuchten Parameters
Average	0.0165	0.011	

Water Retention Plots

Figure 4.1, Control G1 Regression and Raw Data Points (Rows) Removed: [9, 10, 11, 20, 21, 7, 22,6]

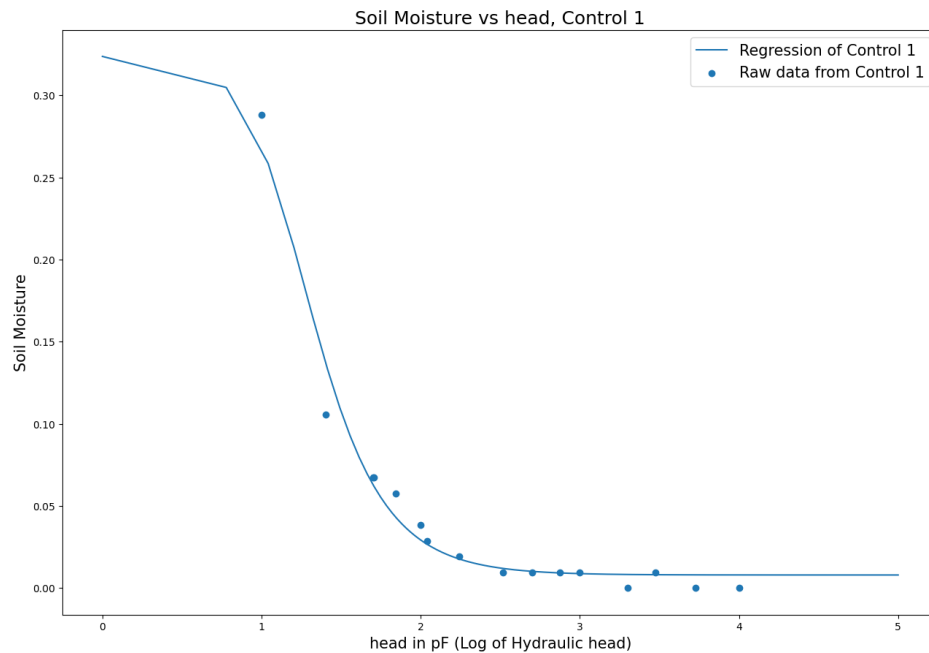


Figure 4.2, Control G2 Regression and Raw Data Points (Rows) Removed: [23, 0, 3, 4, 20, 6, 7, 17, 16]

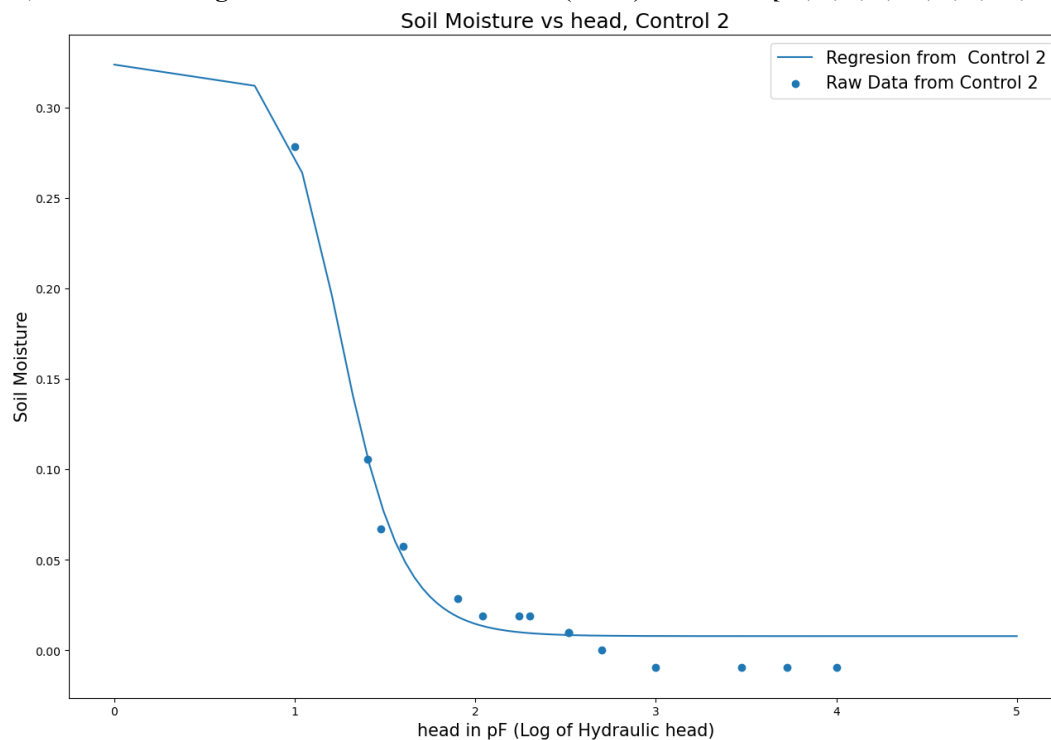


Figure 4.3, Control H1 Regression and Raw Data Points (Rows) Removed: [23, 11, 4, 20, 8, 22, 21, 19]

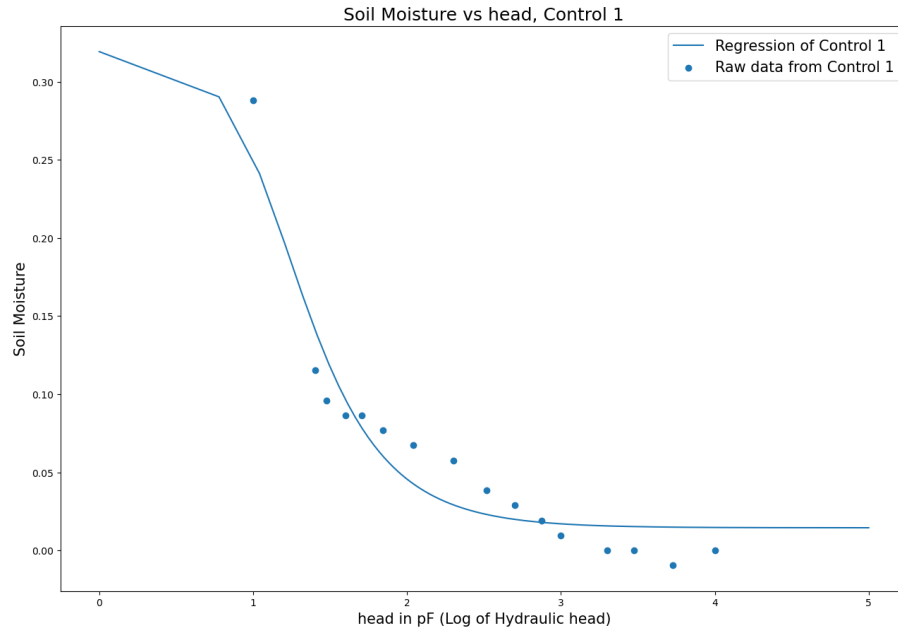
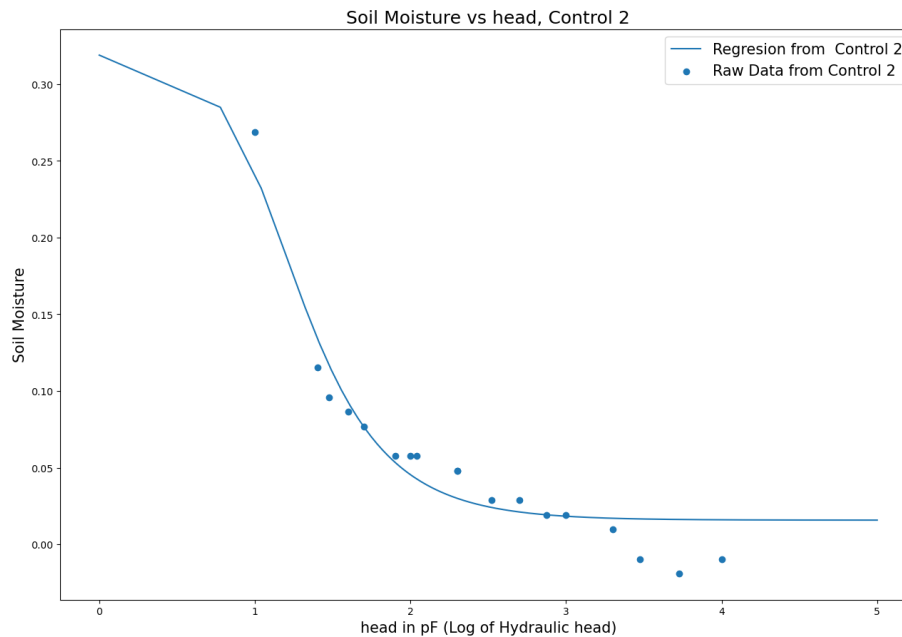


Figure 4.4, Control H2 Regression and Raw Data Points (Rows) Removed: [0, 3, 4, 8, 5, 19]



Section 5 HydraProbe Factory Calibration Validation and Custom Calibration

Section 5a, Calibration Assessment of Sample G297541

Table 5.1, Custom Calibration Statistics for Sample G297541 Rows Removed [26]

Algorithm	Algorithm Type	R Squared	RMSE	Mean Error	Remarks
1	Factory Calibration General Soil	0.853	0.05	-0.043	N= 26
2	Third Order Polynomial Custom Calibration	0.995	0.005	2.164e-16	N = 26
3	Exponential Custom Calibration	0.85	0.0267	-1.417e-16	N=26

Table 5.2, Custom Calibrations for Sample G297541

Algorithm	Equation
1	$\theta = 0.109\sqrt{\varepsilon_r} - 0.179$ (Factory Calibration)
2	$\theta = -0.218 + 0.0997\varepsilon_r - 0.00674\varepsilon_r^2 + 0.0001475\varepsilon_r^3$
3	$\theta = 0.104\sqrt{\varepsilon_r} - 0.1258$

Section 5b, Calibration Assessment of Sample H297546

Table 5.3, Custom Calibration Statistics for Sample H297546, Rows Removed; [None]

Algorithm	Algorithm Type	R Squared	RMSE	Mean Error	Remarks
1	Factory Calibration General Soil	0.911	0.022	-0.0132	N= 27
2	Third Order Polynomial Custom Calibration	0.99	0.006	-1.9711e-16	N=27
2	Exponential Custom Calibration	0.909	0.018	-3.13e-17	N = 27

Table 5.4, Custom Calibrations for Sample H297546

Algorithm	Equation
1	$\theta = 0.109\sqrt{\varepsilon_r} - 0.179$ (Factory Calibration)
2	$\theta = -0.210 + 0.0843\varepsilon_r - 0.00546\varepsilon_r^2 + 0.0001229\varepsilon_r^3$
3	$\theta = 0.1057\sqrt{\varepsilon_r} - 0.158$

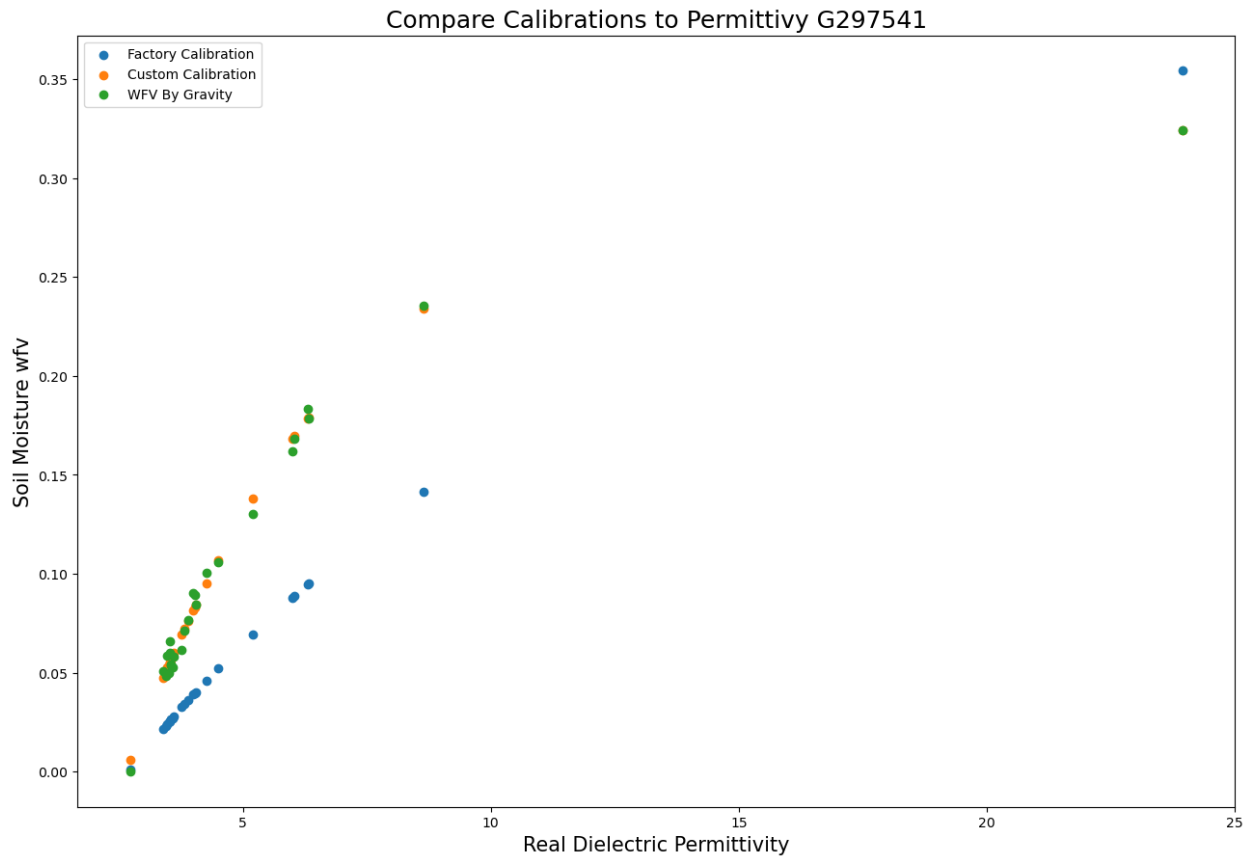
Appendix 1

Appendix 1 Raw Data			
G297541	Sample in a Travel Assy	401cc	Undisturbed/ partially constructed

Table A1, Custom Calibration Data

Real Dielectric (Measured)	wfv HydraProbe Factory (Calculated)	wfv by Gravity (Measured)	Pressure_Bar_(bar) (Measured)	wfv Custom Calibration (Calculated)
2.728	0	0	102000	0.00618212
23.942	0.354	0.32398007	0.0001	0.3239996
6.3	0.096	0.18331436	0.0508	0.17847486
6.03	0.089	0.16834992	0.0508	0.16948694
6.32	0.095	0.17832621	0.0254	0.17911746
8.636	0.141	0.23568989	0.01	0.23398661
6	0.088	0.16186533	0.07	0.16845189
5.2	0.07	0.13044001	0.1	0.13805777
3.52	0.025	0.06010716	0.33	0.0551643
3.9	0.036	0.07656804	0.2	0.07632758
3.52	0.026	0.06609293	0.5	0.0551643
4.05	0.04	0.08454907	0.175	0.08427961
4.5	0.053	0.1059981	0.03	0.10680976
4.26	0.046	0.10051114	0.04	0.09503851
4.03	0.04	0.08953722	0.08	0.08323228
3.998	0.039	0.09053485	0.11	0.08154831
3.5	0.025	0.04963205	10	0.05400935
3.445	0.023	0.04863442	5.35	0.05081178
3.547	0.026	0.05412138	3	0.0567169
3.556	0.027	0.05362257	2	0.05723275
3.476	0.024	0.05861071	0.75	0.05261792
3.579	0.027	0.05262494	1	0.05854724
3.3898	0.022	0.05062968	0.5	0.04757081
3.829	0.034	0.07108108	0.1	0.07248501
3.604	0.028	0.0581119	0.33	0.05996983
3.77	0.033	0.0616036	0.2	0.06925315

Figure A1. Soil moisture vs. Real Dielectric Permittivity



Appendix 2

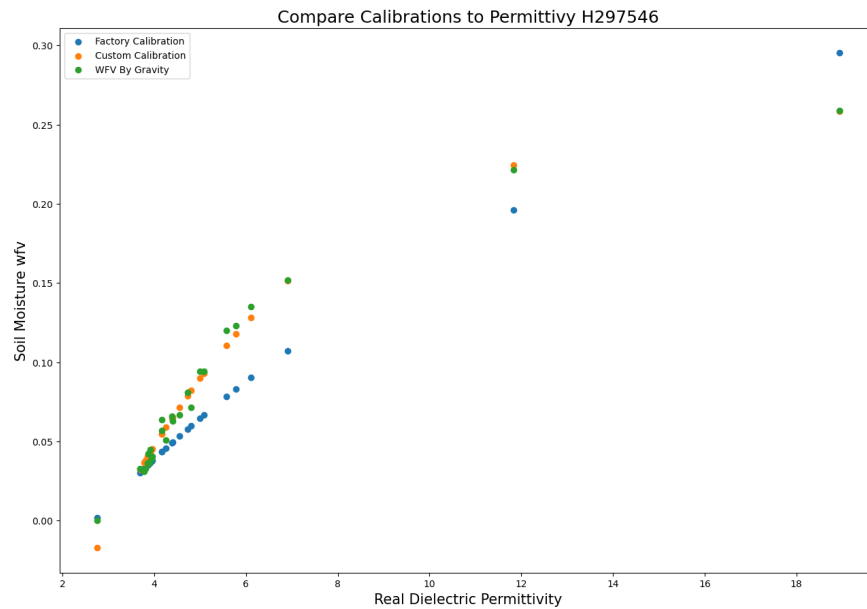
Appendix 2 Raw Data			
H297546	Sample in a Travel Assy	401cc	Undisturbed/ partially constructed

Table A2, Custom Calibration Data

Real Dielectric (Measured)	wfv HydraProbe Factory (Calculated)	wfv by Gravity (Measured)	Pressure_Bar_(bar) (Measured)	wfv Custom Calibration (Calculated)
2.75	0	0	102000	-0.0172137
18.935	0.295	0.25888477	0.0001	0.25846283
6.1	0.092	0.13517875	0.0508	0.12822397
5.78	0.083	0.1232072	0.0508	0.11789156
6.9	0.107	0.15163963	0.0254	0.15122199
11.83	0.196	0.22147367	0.01	0.22462733

5.57	0.078	0.12021431	0.07	0.11074081
5	0.066	0.09427596	0.1	0.08979313
3.86	0.035	0.04239924	0.33	0.04069214
4.16	0.043	0.05686486	0.2	0.05458858
3.91	0.037	0.04489331	0.5	0.04305826
4.4	0.05	0.06285064	0.175	0.06519469
5.08	0.067	0.09427596	0.03	0.09287217
4.72	0.058	0.08080796	0.04	0.07864846
4.382	0.049	0.06584352	0.08	0.06441473
4.546	0.053	0.06684115	0.11	0.07142926
3.8	0.033	0.03292176	10	0.03782608
3.772	0.033	0.0309265	5.35	0.03647857
3.887	0.036	0.03641346	3	0.04197235
3.9	0.036	0.03691228	2	0.04258665
3.944	0.037	0.03940635	0.75	0.04465571
3.857	0.035	0.03641346	1	0.04054953
3.684	0.03	0.03292176	0.5	0.03220183
4.255	0.046	0.05087909	0.1	0.05884048
3.96	0.038	0.04040398	0.333	0.04540423
4.165	0.043	0.06384827	0.2	0.05481413
4.804	0.06	0.07133049	0.05	0.08205253

Figure A2. Soil moisture vs. Real Dielectric Permittivity



Appendix 3 From Retention Curves

Appendix 3 Raw Data			
Sample	Water retention raw		
G_control1	Data		
G_control2			

Table A3, Retention Curve Data

Row #	Pressure Bar (bar)	run_time	G1_wfv_by_grav	G2_wfv_by_grav	Pressure(- cm)
0	0.0508	2.34791667	0.06721075	0.07681229	-50.8
1	0.0254	8.02741389	0.1056169	0.1056169	-25.4
2	0.01	6.39936111	0.28804609	0.27844455	-10
3	0.07	7.82684167	0.05760922	0.05760922	-70
4	0.1	24.0001056	0.03840614	0.04800768	-100
5	0.33	49.7277111	0.00960154	0.00960154	-330
6	0.2	23.5838889	0.02880461	0.02880461	-200
7	0.5	71.8031861	-8.53E-17	-0.0096015	-500
8	0.175	22.640175	0.01920307	0.01920307	-175
9	0.03	7.99634167	0.06721075	0.06721075	-30
10	0.04	6.07867778	0.04800768	0.05760922	-40
11	0.08	7.29249722	0.01920307	0.02880461	-80
12	0.11	5.57332222	0.02880461	0.01920307	-110
13	10	239.944497	-8.53E-17	-0.0096015	-10000
14	5.35	307.226225	-8.53E-17	-0.0096015	-5350
15	3	140.402686	0.00960154	-0.0096015	-3000
16	2	188.582364	-8.53E-17	0.01920307	-2000
17	0.75	140.692103	0.00960154	0.01920307	-750
18	1	161.933483	0.00960154	-0.0096015	-1000
19	0.5	196.793147	0.00960154	-8.53E-17	-500
20	0.1	24.8407944	0.04800768	0.03840614	-100
21	0.33	70.1022639	0.02880461	0.00960154	-330
22	0.2	73.1083306	0.02880461	0.01920307	-200
23	0.05	24.0141556	0.06721075	0.05760922	-50

Appendix 3 Raw Data			
Sample	Water retention raw		
H_control1	Data		
H_control2			

Table A4, Retention Curve Data

Row#	Pressure Bar (bar)	run_time	H1_wfv_by_grav	H2_wfv_by_grav	Pressure(- cm)
0	0.0508	2.34791667	0.08641383	0.09601536	-50.8
1	0.0254	8.04072222	0.11521843	0.11521843	-25.4
2	0.01	6.39050556	0.28804609	0.26884301	-10
3	0.07	7.82242222	0.07681229	0.07681229	-70
4	0.1	24.0001056	0.05760922	0.04800768	-100
5	0.33	49.7244917	0.03840614	0.03840614	-330
6	0.2	23.5801611	0.05760922	0.04800768	-200
7	0.5	71.8032694	0.02880461	0.02880461	-500
8	0.175	22.6362611	0.04800768	0.05760922	-175
9	0.03	7.97417222	0.09601536	0.09601536	-30
10	0.04	6.07404167	0.08641383	0.08641383	-40
11	0.08	7.20155556	0.05760922	0.05760922	-80
12	0.11	5.58921667	0.06721075	0.05760922	-110
13	10	238.510067	-8.53E-17	-0.0096015	-10000
14	5.35	308.173989	-0.0096015	-0.0192031	-5350
15	3	140.424525	-8.53E-17	-0.0096015	-3000
16	2	188.608825	-8.53E-17	0.00960154	-2000
17	0.75	139.578967	0.01920307	0.01920307	-750
18	1	162.130875	0.00960154	0.01920307	-1000
19	0.5	196.794722	0.00960154	0.00960154	-500
20	0.1	24.8527	0.04800768	0.05760922	-100
21	0.333	68.0814778	0.01920307	0.02880461	-333
22	0.2	73.10705	0.02880461	0.04800768	-200
23	0.05	24.0111833	0.06721075	0.07681229	-50

Appendix 4 Photos of Samples



Sample G297541 and controls before drying.



Sample H297546 with controls before drying.



Sample G297541 after drying.



Sample H297546 after drying.